

Linear program (solved exactly with HiGHS)

Decision variable: $x_{r,t} \geq 0$ — flexible load run in region r at hour t

Objective: $\min_{x, M} \sum_{r,t} x_{r,t} \text{CI}_{r,t} (1 - \alpha \text{CFE}_{r,t}) + \eta M$

C1

Demand

C2

Latency

C3

Capacity

C4

Geographic
cap

C5

Fairness

C6

Ramp
rate

C7

Dynamic
range

subject to seven operational constraints